



Republic of the Philippines

University of Cabuyao

(Pamantasan Ng Cabuyao)

College of Computing Studies

Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna, Philippines 4025



MIDTERM EXAMINATION

ITP105- NETWORKING AND COMMUNICATION 2

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I. Hands-on. (60 pts)

Problem:

A local school wants to improve its network infrastructure to support different departments: Admission, Accounting, and Registrar. Each department requires its own VLAN to ensure secure communication and efficient resource management. The school has decided to implement inter-VLAN routing to allow communication between these VLANs while using static routing or dynamic RIP routing protocol for connecting to the main office.

Technical Specifications:

1. Design a network topology showing the router, switch, and devices for each VLAN. Include the physical connections and assigned network addresses.
2. Configure VLANs on the Switch.
3. Change the Hostname of each switch and router. Include your surname in the new hostname.
4. Set Up Inter-VLAN Routing. Configure the router with sub-interfaces for each VLAN and assign the appropriate IP addresses.
5. Implement Static Routing. If the router has multiple paths, configure static routes to ensure traffic flows correctly between the VLANs.
6. Test the connections.

Deliverables:

1. Create a video tutorial that explains the concept of inter-VLAN, and Static Routing.
2. Demonstrate how to perform the technical specifications in the Cisco Packet Tracer. Make sure that you are visible in the video.
3. Limit your recordings to a maximum of 5 minutes.
4. Upload your video demonstration to YouTube.com.
5. PASTE YOUR YOUTUBE LINK HERE:

6. Supply the required screenshots below.



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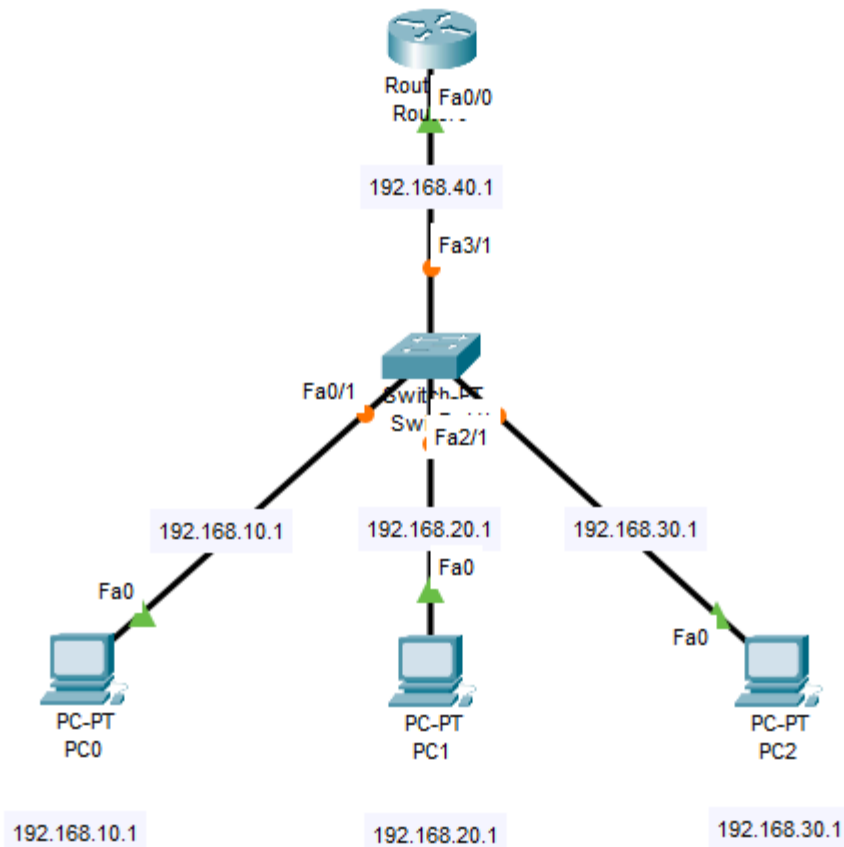
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Press RETURN to get started!

```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet2/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
```

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Orcasitas_S1
Orcasitas_S1(config)#
Orcasitas_S1(config)#
Orcasitas_S1(config)#
Orcasitas_S1(config)#vlan 10
Orcasitas_S1(config-vlan)#name Admission
Orcasitas_S1(config-vlan)#exi
Orcasitas_S1(config)#vlan 20
Orcasitas_S1(config-vlan)#name Accounting
Orcasitas_S1(config-vlan)#exi
Orcasitas_S1(config)#vlan 30
Orcasitas_S1(config-vlan)#name Registrar
Orcasitas_S1(config-vlan)#exi
Orcasitas_S1(config)#exi
Orcasitas_S1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy

Paste



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```
Orcasitas_Sl(config)#vlan 30
Orcasitas_Sl(config-vlan)#name Registrar
Orcasitas_Sl(config-vlan)#exi
Orcasitas_Sl(config)#exi
Orcasitas_Sl#
%SYS-5-CONFIG_I: Configured from console by console

Orcasitas_Sl#config t
Enter configuration commands, one per line. End with CNTL/Z.
Orcasitas_Sl(config)#int fa0/1
Orcasitas_Sl(config-if)#switchport mode access
Orcasitas_Sl(config-if)#switchport access vlan 10
Orcasitas_Sl(config-if)#exi
Orcasitas_Sl(config)#
Orcasitas_Sl(config)#int fa1/1
Orcasitas_Sl(config-if)#switchport mode access
Orcasitas_Sl(config-if)#switchport access vlan 20
Orcasitas_Sl(config-if)#exi
Orcasitas_Sl(config)#
Orcasitas_Sl(config)#int fa2/1
Orcasitas_Sl(config-if)#switchport mode access
Orcasitas_Sl(config-if)#switchport access vlan 30
Orcasitas_Sl(config-if)#exi
Orcasitas_Sl(config)#exi
Orcasitas_Sl#
%SYS-5-CONFIG_I: Configured from console by console

Orcasitas_Sl#int g0/1|
^
% Invalid input detected at '^' marker.

Orcasitas_Sl#interface GigabitEthernet0/1
^
% Invalid input detected at '^' marker.

Orcasitas_Sl#interface fa3/1
^
```

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```
Orcasitas_Sl#config t
Enter configuration commands, one per line. End with CNTL/Z.
Orcasitas_Sl(config)#int fa3/1
Orcasitas_Sl(config-if)#switchport mode trunk
Orcasitas_Sl(config-if)#exi
Orcasitas_Sl(config)#exi
Orcasitas_Sl#
%SYS-5-CONFIG_I: Configured from console by console
```

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```
Orcasitas_Rl>en
Orcasitas_Rl#config t
Enter configuration commands, one per line. End with CNTL/Z.
Orcasitas_Rl(config)#int g0/0.10
%Invalid interface type and number
Orcasitas_Rl(config)#int fa0/0.10
Orcasitas_Rl(config-subif)#encapsulation dot1Q 10
Orcasitas_Rl(config-subif)#ip add 192.168.10.2 255.255.255.0
Orcasitas_Rl(config-subif)#exi
Orcasitas_Rl(config)#
Orcasitas_Rl(config)#int fa0/0.20
Orcasitas_Rl(config-subif)#encapsulation dot1Q 20
Orcasitas_Rl(config-subif)#ip add 192.168.20.2 255.255.255.0
Orcasitas_Rl(config-subif)#exi
Orcasitas_Rl(config)#
Orcasitas_Rl(config)#int fa0/0.30
Orcasitas_Rl(config-subif)#encapsulation dot1Q 30
Orcasitas_Rl(config-subif)#ip add 192.168.30.2 255.255.255.0
Orcasitas_Rl(config-subif)#exi
Orcasitas_Rl(config)#
Orcasitas_Rl(config)#ip route 192.168.40.0 255.255.255.0 [next_hop_ip]
^
```



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```
Orcasitas_R1(config-if)#ip add 192.168.40.1 255.255.255.0
Orcasitas_R1(config-if)#no shut
```

```
Orcasitas_R1(config-if)#ip route 192.168.40.0 255.255.255.0 192.168.30.1
Orcasitas_R1(config)#exi
Orcasitas_R1#
%SYS-5-CONFIG_I: Configured from console by console
exi
```

```
Orcasitas_R1>en
Orcasitas_R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Orcasitas_R1(config)#router rip
Orcasitas_R1(config-router)#version 2
Orcasitas_R1(config-router)#network 192.168.10.0
Orcasitas_R1(config-router)#network 192.168.20.0
Orcasitas_R1(config-router)#network 192.168.30.0
Orcasitas_R1(config-router)#exi
Orcasitas_R1(config)#exi
Orcasitas_R1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Orcasitas_S1>en
Orcasitas_S1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Orcasitas_S1(config)#int fa4/1
Orcasitas_S1(config-if)#switchport trunk mode
^
% Invalid input detected at '^' marker.
```

```
Orcasitas_S1(config-if)#switchport mode trunk
Orcasitas_S1(config-if)#exi
Orcasitas_S1(config)#
```